



AMANAI LAB
AI EDUCATION &
MENTORSHIP

LIVE COHORT · 2026

GenAI & Agentic AI Complete Syllabus

A complete, industry-mapped curriculum covering LLM foundations, RAG, Agentic AI, MCP/A2A protocols, and production deployment — built around real, end-to-end production-grade projects, not toy notebooks.

Enterprise RAG Systems

Multi-Agent Orchestration

MCP & A2A Protocols

Production LLMops

CURRICULUM FLOW

Foundations



Generative Models



RAG Systems



Agentic AI



MCP/A2A



LLMOps



5 Projects



Capstone

Built by a practicing AI/ML Engineer — every module is grounded in what's actually shipped in production, not just theory.

Master Topic Map

Every concept mapped into the curriculum — built to leave no gap for what's actually asked in 2026 GenAI interviews and used in production systems.

● Foundations

Python for AI · NLP basics · Word embeddings (Word2Vec/GloVe) · Transformer architecture & self-attention · Encoder/Decoder models (BERT vs GPT) · Tokenizers (BPE, SentencePiece)

● Generative Model Core

Autoencoders & VAEs · GANs · Diffusion models (Stable Diffusion) · CLIP (contrastive image-text pretraining) · Vision Transformers (ViT) · Multimodal embeddings

● LLM Core Concepts

Pretraining, SFT, RLHF/DPO/ORPO/KTO · Decoding strategies (top-k, top-p, temperature) · Hallucination causes & fixes · Structured outputs (JSON mode, Pydantic/Instructor)

● Prompting & Context Engineering

Zero/few-shot, CoT, ToT, ReAct, self-consistency · Context engineering (beyond prompting) · Prompt injection risks · Prompt versioning & management

● Reasoning Models & Test-Time Compute

Reasoning tokens & chain-of-thought at inference · Test-time compute scaling · o1/o3-style reasoning models · When reasoning models beat standard LLMs

● Fine-Tuning & Model Optimization

LoRA, QLoRA, PEFT · Full fine-tuning vs adapters · Quantization (INT8/INT4) · Knowledge distillation · Synthetic data generation for fine-tuning

● RAG — Core to Advanced

Embedding models & similarity metrics · Vector DBs (FAISS, Pinecone, Weaviate, Chroma) · Index types (HNSW, IVF) · Chunking · Hybrid search & re-ranking · HyDE, query rewriting, Graph RAG

● Agentic RAG & Long-Context Debate

Agents that retrieve autonomously vs static pipelines · RAG vs long-context windows — when each wins · Hybrid RAG + long-context architectures

● Document Intelligence & Multimodal RAG

Parsing PDFs, tables & scanned docs · Layout-aware extraction · Image + text multimodal retrieval · Enterprise document pipelines

● RAG Evaluation & Reliability

RAGAS, DeepEval, TruLens · Faithfulness, relevance & context precision/recall · Hallucination mitigation · Citation grounding

● Agentic AI

Agent architecture & ReAct pattern · Planning & task decomposition · Tool/function calling · Guardrails & safety (loop prevention, cost caps, human-in-the-loop)

● Agent Memory & Context Systems

Short/long-term memory · Episodic & semantic memory · Memory frameworks (Mem0, Zep-style) · Context window management at scale

● Frameworks & Protocols

LangChain · LangGraph (state graphs, checkpointing) · CrewAI · AutoGen · LlamaIndex · MCP (Model Context Protocol) · A2A (Agent-to-Agent Protocol)

● Mixture of Experts & Architecture Trends

MoE routing & sparse activation · Why frontier models use MoE · Dense vs sparse model tradeoffs · Frontier model landscape awareness (GPT, Claude, Gemini, Llama, DeepSeek families)

● Model Routing, Caching & Cost Engineering

Prompt caching · Model cascading/routing (cheap → expensive) · Inference cost optimization · Latency vs accuracy tradeoffs at scale

● LLMops, Security & Voice Agents

Model serving & deployment · Observability (LangSmith) · Guardrails & PII redaction · Whisper STT/TTS · Computer-use / browser agents

WHY THIS CURRICULUM IS DIFFERENT

01

Production-First, Not Notebook-First

Every topic ties directly into a deployed, production-grade project — not an isolated Jupyter demo that never ships.

03

Protocol-Level Depth (MCP/A2A)

Most courses stop at LangChain basics. This one goes into the protocols powering next-gen enterprise agent systems.

02

Built by a Practicing Engineer

Curriculum shaped by real enterprise pipelines (Fortune-500 scale data & agent systems), not repackaged textbook content.

04

Career Outcome Baked In

Resume, LinkedIn and a personal roadmap are part of the curriculum — not an afterthought bolted on at the end.

Flagship Projects — Built Live, End-to-End

Five production-grade projects, each mirroring a real enterprise use case — not toy demos. Every project is discussed for architecture decisions, tradeoffs and how to defend it in an interview.

PROJECT 01

Enterprise RAG Support Assistant

A multi-source document Q&A assistant for customer support — ingests policy docs, PDFs & FAQs, retrieves with hybrid search + re-ranking, and answers with grounded citations.

LangChain / LlamaIndex

Pinecone / Weaviate

Cross-Encoder Re-ranker

FastAPI

RAGAS Evaluation

Project discussion: Why hybrid search beats pure dense retrieval, how citation grounding reduces hallucination, and where this architecture breaks at scale (and how to fix it).

PROJECT 02

Multi-Agent Research & Report Generation System

A supervisor agent decomposes a research task, delegates to specialist sub-agents (search, summarize, fact-check), and compiles a structured report.

LangGraph

Multi-Agent Orchestration

Web Search Tool Integration

Agent Memory

Project discussion: Supervisor-worker vs peer-to-peer agent patterns, handling agent failure/retry loops, and evaluating multi-agent output quality.

PROJECT 03

Autonomous Business Workflow Agent (MCP-Integrated)

An agent that automates a real business process — e.g. support-ticket triage and CRM updates — by connecting to tools through MCP with human-in-the-loop approval for risky actions.

MCP Servers

Function Calling

Guardrails

Human-in-the-Loop

Project discussion: Why MCP standardizes tool access across agents, designing safe autonomy boundaries, and rollback strategies when an agent acts incorrectly.

PROJECT 04

Voice-Enabled Real-Time AI Agent

A voice-driven agent that listens (Whisper STT), reasons + calls tools, and responds naturally (TTS) — simulating a real-time voice assistant use case.

Whisper STT

TTS Integration

Real-Time Pipelines

Tool Calling

Project discussion: Latency budgets for real-time voice UX, streaming responses, and where voice agents fail in production (noise, interruptions, turn-taking).

PROJECT 05 — CAPSTONE

End-to-End Production GenAI Application

The final capstone — combines RAG + Agents + protocol integration into one deployed system, with monitoring, observability, guardrails and cost tracking, mirroring what a real AI Engineer ships.

Full-Stack RAG + Agents

Azure / AWS Deployment

LangSmith Observability

Semantic Caching

Guardrails AI

Project discussion: Full dev-to-production lifecycle, cost optimization decisions, and how to present this as your flagship interview project — code-reviewed and GitHub-pushed.

Bootcamp — GenAI & Agentic AI

A hands-on, production-first curriculum — every module builds toward one of the five flagship projects above.

- Hands-on heavy
- 5 production projects
- Ends with deployed capstone
- Includes career layer

MODULE — Foundations & Core NLP

1	Python for AI + NLP & Embedding Basics	Python essentials, tokenization, Word2Vec vs contextual embeddings
2	Transformer Architecture Deep-Dive	Self-attention, positional encoding, Encoder vs Decoder (BERT vs GPT)
3	How LLMs Are Trained	Pretraining, SFT, RLHF/DPO, decoding strategies — hands-on first API calls

MODULE — Prompting, Context Engineering & Generative Foundations

4	Prompt & Context Engineering Mastery	CoT, ToT, ReAct, self-consistency, structured outputs (Pydantic/Instructor)
5	Generative Model Core	Autoencoders, VAEs, GANs, Diffusion models, CLIP & Vision Transformers
6	Model Landscape & Fine-Tuning	Open vs closed LLMs, LoRA/QLoRA/PEFT, DPO/ORPO/KTO, quantization & distillation, MoE architecture basics

MODULE — Embeddings, Vector Search & Document Intelligence

7	Embedding Models & Similarity Metrics	Choosing embedding models, cosine/dot/euclidean similarity, HNSW/IVF indexes
8	Vector Databases	FAISS, Pinecone, Weaviate, Chroma — setup, comparison & access control
9	Document Intelligence & Multimodal Retrieval	Parsing PDFs/tables, layout-aware extraction, image+text retrieval
★	Project Build — Enterprise Semantic Search	Hands-on build combining embeddings + vector DB + document pipeline

MODULE — RAG Systems: Core to Advanced

10	RAG Architecture End-to-End	Indexing → retrieval → generation, chunking strategies
11	Retrieval Techniques	Dense vs sparse (BM25), hybrid search, re-ranking with cross-encoders
12	Advanced RAG Patterns	HyDE, query rewriting, multi-query, parent-child chunking, Graph RAG, Agentic RAG
13	RAG Evaluation Frameworks	RAGAS, DeepEval, TruLens, faithfulness & context precision/recall
14	RAG vs Long-Context — The 2026 Debate	When long-context windows replace RAG, hybrid architectures, cost tradeoffs
★	Project Build — Enterprise RAG Support Assistant	Full project: hybrid search, re-ranking, citation grounding

MODULE — Production RAG, Reliability & Security

15	Production RAG Challenges	Latency, cost, stale-data handling — hands-on optimization
16	Guardrails & Hallucination Mitigation	Guardrails AI / NeMo Guardrails, PII redaction
17	Security for GenAI Systems	Prompt injection defense, access control on vector DBs, data leakage prevention

MODULE — Agentic AI Foundations

18	What Makes a System Agentic	Agent architecture, ReAct pattern, planning & task decomposition
19	Tool Calling & Structured Outputs	Function calling, schema-validated structured responses
20	Agent Memory & Context Systems	Short/long-term memory, episodic & semantic memory, Mem0/Zep-style systems

MODULE — Frameworks, Multi-Agent Systems & Protocols

21	LangGraph Deep-Dive	State graphs, cyclic workflows, checkpointing
22	Multi-Agent Orchestration	CrewAI / AutoGen, supervisor-worker & hierarchical patterns
23	MCP & A2A Protocols	Model Context Protocol and Agent-to-Agent communication standards
★	Project Build — Multi-Agent Research System	Supervisor + specialist sub-agents, report generation
★	Project Build — Autonomous Workflow Agent (MCP)	Real business process automation with human-in-the-loop

MODULE — Voice & Real-Time Agents

24	Speech Integration in Agent Pipelines	Whisper (STT), TTS integration, real-time turn-taking
★	Project Build — Voice-Enabled Real-Time Agent	End-to-end voice agent with tool calling

MODULE — LLMops, Deployment & Evaluation

25	Model Serving & Deployment	vLLM, TGI, deployment on Azure/AWS/GCP, streaming responses
26	Observability & Cost Governance	LangSmith, OpenTelemetry, semantic caching, token budgeting
27	Model Routing, Prompt Caching & Cost Engineering	Model cascading (cheap → expensive), prompt caching, inference cost optimization
28	Agent & System Evaluation	LLM-as-judge, agent benchmarking, A/B testing prompts & models

MODULE — Capstone & Career Launch

★	Capstone Project — Full Production GenAI App	RAG + Agents + Protocol + Deployment + Monitoring, code-reviewed
29	Capstone Presentations	Present & defend your architecture decisions

✦ INCLUDED WITH EVERY BOOTCAMP SEAT

5 Production Projects

Portfolio-ready, GitHub-pushed, code-reviewed

LinkedIn Optimization

Headline, About section, project showcase

Resume Review

Tailored for GenAI / AI Engineer roles

Personal AI Roadmap

Individualized next-skills & career plan

WHO THIS BOOTCAMP IS FOR

- ✓ Software engineers & data professionals moving into GenAI/AI Engineering roles
- ✓ ML/DL practitioners wanting production-grade RAG & Agentic AI depth
- ✓ Anyone targeting AI Engineer, GenAI Developer or Applied AI roles in 2026

PREREQUISITES

- ✓ Working knowledge of Python (functions, OOP basics)
- ✓ Basic understanding of ML concepts (helpful, not mandatory)
- ✓ A laptop and willingness to build — everything else is taught live



AMANAI LAB

GenAI & Agentic AI Bootcamp — Live Cohort 2026

₹34,999

₹24,999 / seat

Early-bird cohort pricing · Limited seats · EMI options available

ENROLL NOW — SEATS LIMITED

EVERYTHING YOU GET WHEN YOU ENROLL



Full Live Curriculum

Every module in this syllabus, taught live via Zoom Pro



Lifetime Recording Access

Rewatch any class anytime — never miss a concept



5 Production-Grade Projects

Portfolio-ready, GitHub-pushed, code-reviewed by mentor



Deployed Capstone Project

Full RAG + Agents + Protocol system, live-reviewed



Resume Review

Tailored specifically for GenAI / AI Engineer roles



LinkedIn Optimization

Headline, About section & project showcase, reviewed live



Personal AI Career Roadmap

Individualized next-skills & job-target plan



Certificate of Completion

Shareable on LinkedIn once you finish the capstone



Doubt Support & Community

Direct access to mentor for questions through the cohort



Interview-Ready Portfolio

Walk into interviews with 5 real projects to discuss, not theory

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